

# Electrical Machine- II (BT EE403A)



हरियाणा केंद्रीय विश्वविद्यालय  
**CENTRAL UNIVERSITY OF HARYANA**  
NAAC ACCREDITED 'A' GRADE UNIVERSITY

## Unit-IV

### Synchronous Machine

- ❖ Constructional features
- ❖ Cylindrical rotor synchronous machine
- ❖ Generated EMF

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# Alternator

- Synchronous generator or AC generator is a device which converts mechanical power in the form of A.C.
- It works on the principle of ELECTRO MAGNETIC INDUCTION and it is also called as **Alternator**.
- An alternator consists of armature winding and field magnet, but the difference between the alternator and DC generator is that in the DC generator armature rotates and the field system is stationary. This arrangement is the alternator is just reverse of it there the armature is stationary called as stator and field system is rotating called as Rotor.

## For generating EMF, three things are essential:

- 1) Magnetic field
- 2) System of conductors
- 3) Relative motion between those two.

➤ The conductors are mounted on the stators and the field poles are mounted on the Rotor core. Relative motion between the stator conductors and the field is brought about by rotating the field system.

➤ The rotor is coupled mechanically to a suitable prime mover. When the prime mover runs, the rotor core also rotates and the field flux is cut by the stationary stator conductors and emf's are induced in them.

➤ If a load is connected across the stator terminals, electric power would be delivered to it.

## Advantages of Stationary Armature

- The generated power can be easily taken out from the stator.
- There is no possibility of the armature conductors flying off, when the machine runs at high speed since they are housed in the stator slots.
- There is no difficulty in insulating the armature (stationary) winding for very high voltages, i.e., as high as 30000v or more.
- Two slip rings are required for the supply of DC energy required for rotor field excitation.
- Since exciting current is to be supplied at low voltage, there is no difficulty in insulating them.
- Rotating field is comparatively light and can run with high speeds.

## Difference Between Stationary and Rotating Field System

| S.No. | Stationary Field System                          | Rotating Field System                             |
|-------|--------------------------------------------------|---------------------------------------------------|
| 1     | 4 slip rings are required                        | 100 slip rings are required                       |
| 2     | Heavy armature current passes through slip rings | Very low field current passes through slip rings  |
| 3     | More sparking at slip rings                      | No sparking at slip rings                         |
| 4     | Armature supply is taken through slip rings      | Armature supply is taken through fixed connection |
| 5     | Capacity is Limited to 30KVA                     | It can be designed to any capacity                |
| 6     | Voltage is limited to 440v                       | Voltage is up to 33KV is generated                |
| 7     | Lower efficiency                                 | High efficiency                                   |
| 8     | Mare maintenance                                 | Less maintenance                                  |

# Construction

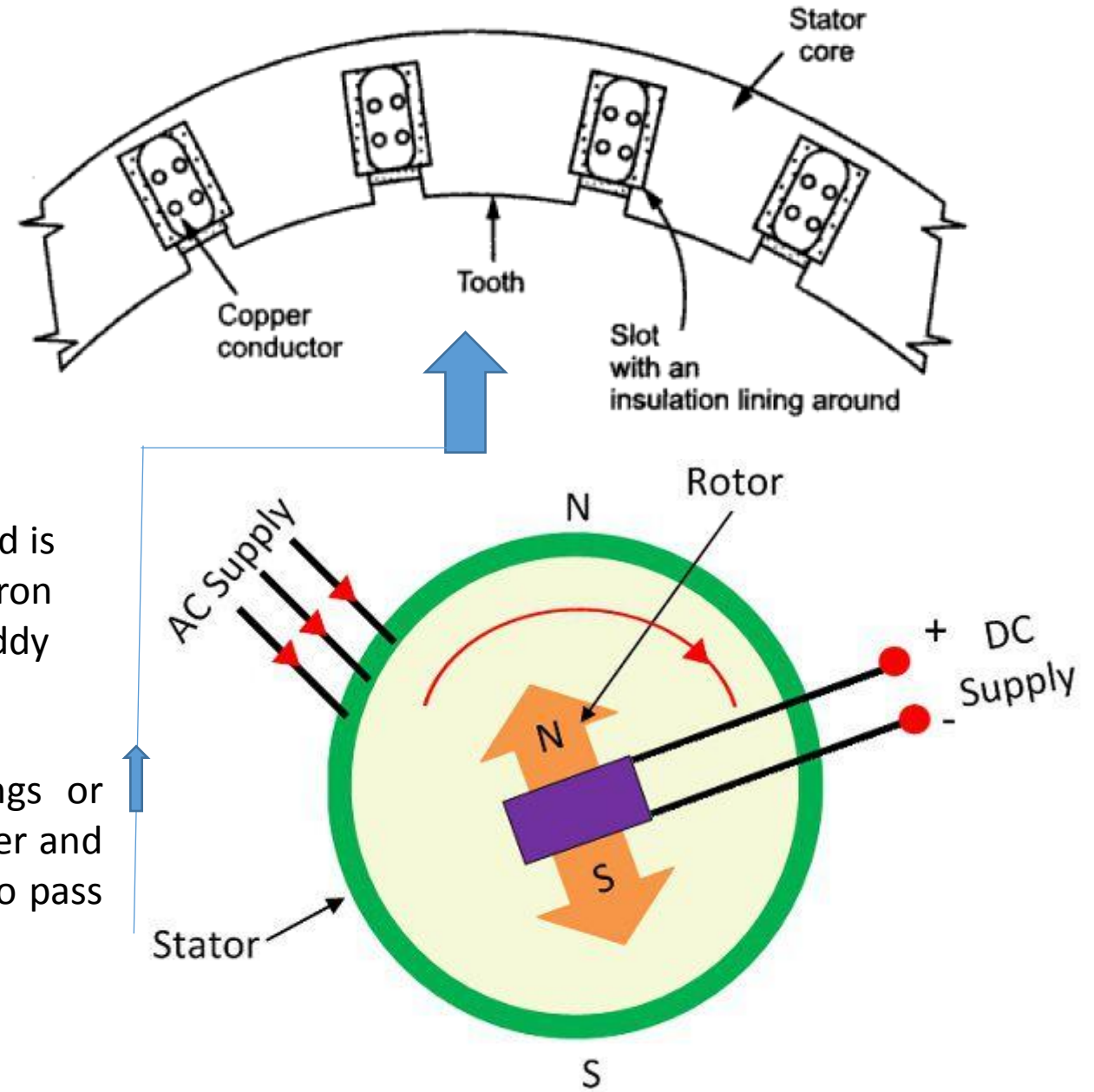
An alternator consists of mainly two parts

1. Stator
2. Rotor

## 1. Stator:-

➤The armature core is supported by the stator frame and is built up of laminations of special magnetic iron or steel iron alloy the core is laminated to minimize the loss due to Eddy currents.

➤The laminations are stamped out in complete rings or segments. The laminations are insulated from each other and have space between them for allowing the cooling air to pass through.



## Stator:-

- The inner periphery of the stator is slotted and copper conductors which are joined to one another constituting armature winding housed in these slots. The other ends of the winding are brought out and are connected to fixed terminal from which the generator power can be taken out.
- The wide open type slot also used in DC machines has the advantage of permitting easy installation of form-wound coils and their easy removal in case of repair but it has the disadvantage of distributing the air gap flux into bunches that produce ripples in the wave of generated EMF.
- The semi closed type slots are better in this respect but do not allow the use of form wound coils.
- The fully closed slots do not disturb the air gap flux but they try to increase the inductance of the windings. The armature conductors have to be threaded through, thereby increasing the initial labour and cost of the winding. Hence, these are rarely used.

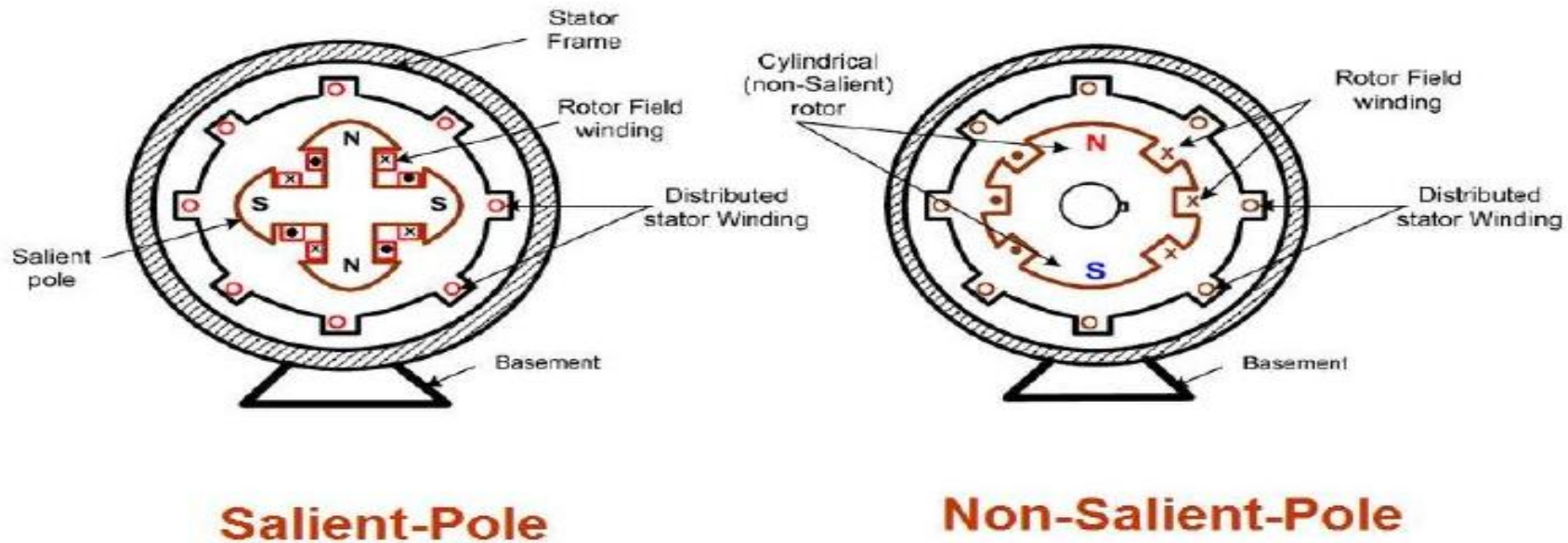
## 2. Rotor

Depending upon the type of application, these are classified into two types

A. Salient-pole or projecting pole type

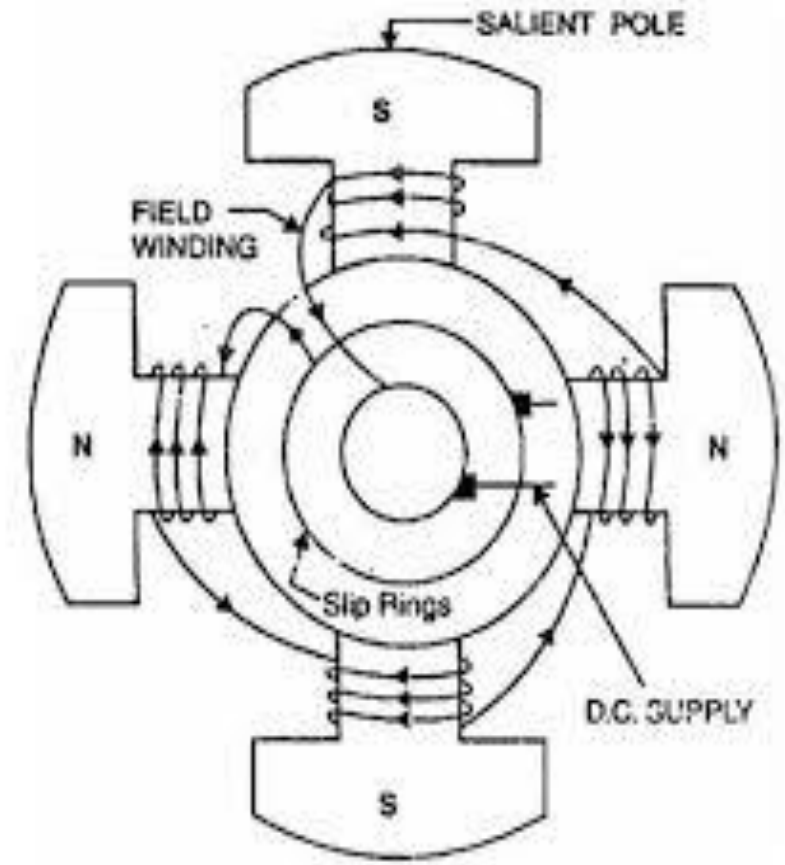
B. Non salient-pole or round rotor or cylindrical rotor

### Salient-Pole **VS** Non-salient-Pole



## Salient-pole or projecting pole type Rotor

- It is used for and medium speed alternators used in hydro and diesel power generating station.
- The poles are made of laminated sheets and fixed to the rotor by dove tail joint.
- Short circuited damper bars are placed in the slots provided on the pole surfaces.
- These are used to prevent hunting and to provide starting torque in synchronous motors.
- The field coils are placed on the poles as shown in the figure



Salient-pole or projecting pole type Rotor

## Salient-pole or projecting pole type Rotor

### Key features:-

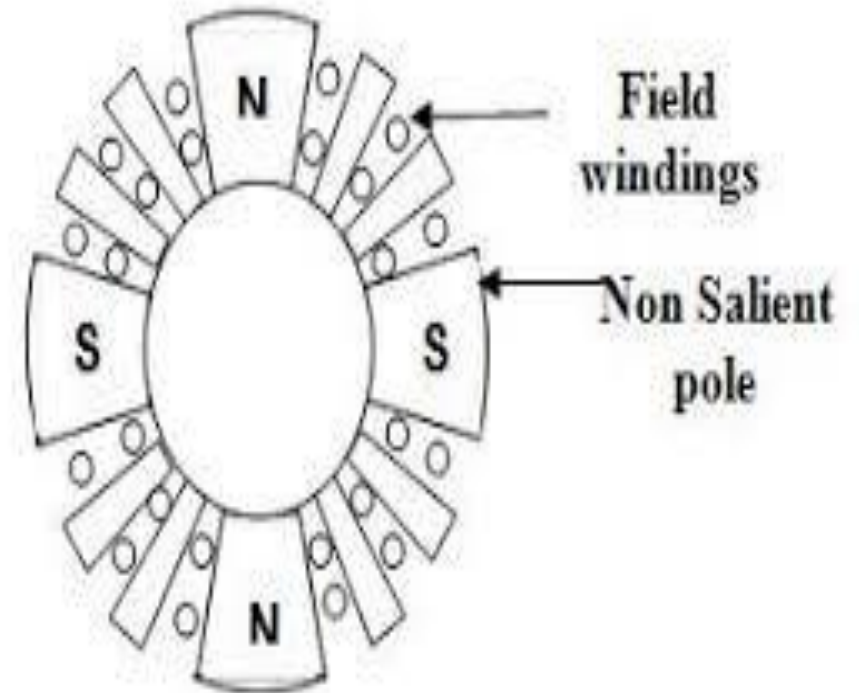
- It has non-uniform air gap.
- The diameter of the rotor is more than of the cylindrical rotor.
- The no. of holes is higher than that of the non salient-pole rotor
- Axial length is less.
- The prime mover speed is less and is driven in hydro turbines
- These generators are used in hydro electric stations so these are called as hydro generators.

## Non-Salient pole type (or) Cylindrical type (or) Round rotor:

➤ Slots are provided in between the poles and these slots are placed with field winding Conductors.

### Features :-

- No. of poles are less when compared to salient pole type.
- Diameter is less
- Axial length is more
- Air gap is uniform
- Prime mover speed is more and is driven in thermal turbines.
- These are used in thermal stations so, these are called as turbo Generators.



## Frequency of the induced EMF:-

- Consider an alternate whose rotor is driver at a constant speed  $N$  rpm.
- Let “ $p$ ” be no. of poles and  $f$  is the frequency of the generated voltage

No. of cycles of the induced EMF per sec

$$\begin{aligned} &= \text{No. of cycles per revolutions} \times \text{no. of revolutions/sec} \\ &= P/2 \times N/60 = PN \times 120. \end{aligned}$$

Frequency  $f = PN/120$  Hz.

For a given alternate, the no. of poles is fixed.

Hence in order to generate power at a specified frequency, the machine is to be run at a definite speed which is termed as synchronous speed.

## NOTES:

**Pole pitch:-** distance between two adjacent opposite main poles by the no. of armature conductors.

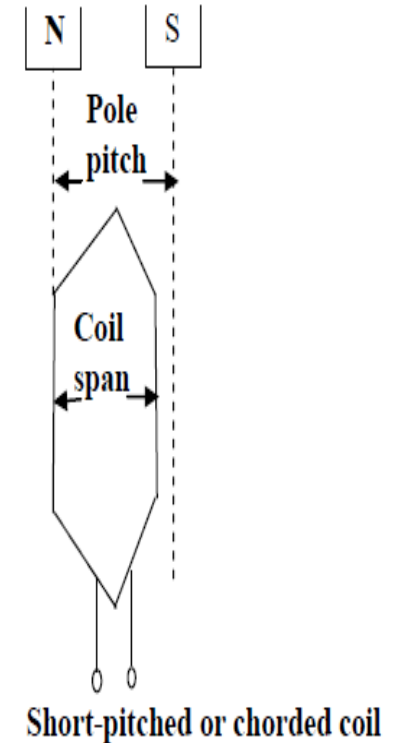
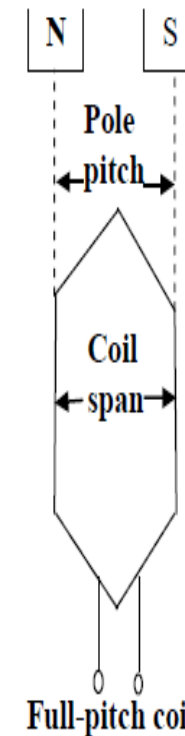
**Coil span:-** distance b/w two coils starting and ending conductors

If the coil span = pole pitch, then full pitch winding (or) integral slot winding.

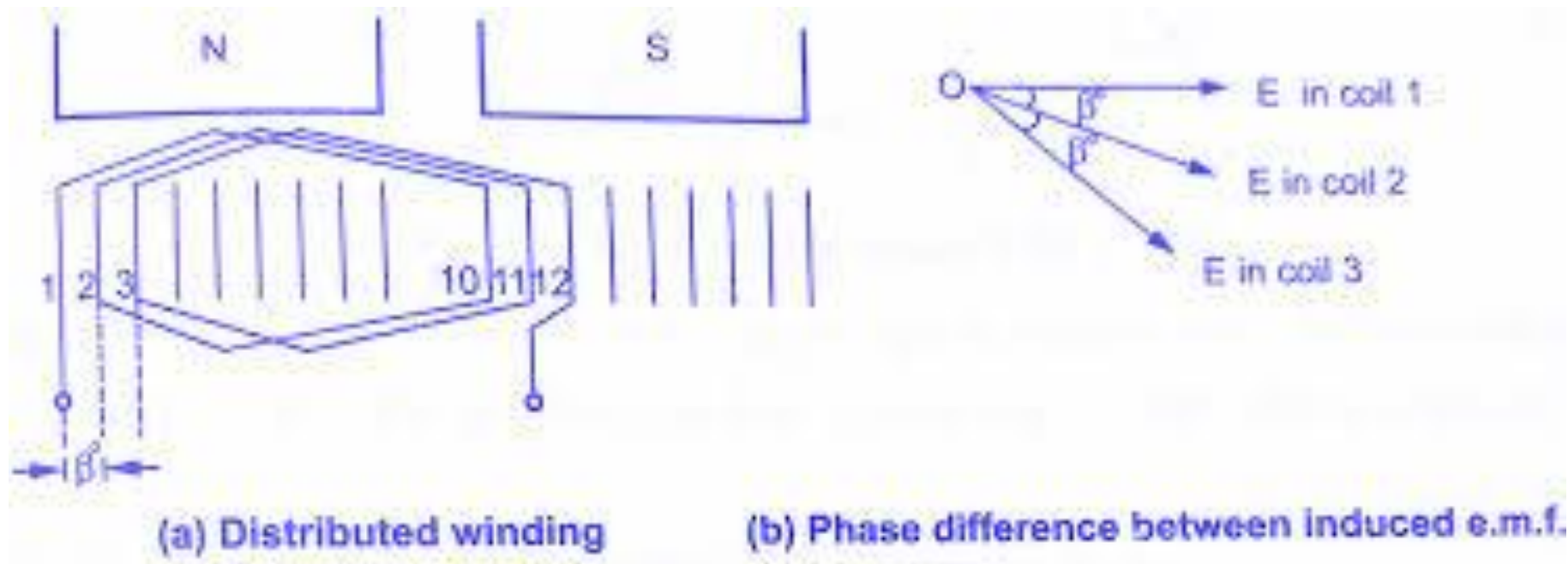
Integral slot (or) full pitch winding and short pitch (or) fractional chorded winding .

The distance between any two conductors is called slot angle ( $\beta$ )

Short pitch angle ( $\alpha$ ) = short chorded slots  $\times$  B



- If the distance b/w two coils sides of a coil, i.e. coil span is equal to one pole pitch, i.e.  $180^\circ$ , it is called as full pitch winding.
- If the distance b/w two coils sides of a coil, i.e. coil span is less than one pole pitch i.e.  $180^\circ$ , it is called short pitch (or) fractional chorded winding.
- If the winding is short pitched by one slot then the short pitch angle  $\alpha$  is equal to slot angle  $\beta$ .
- If it is short pitched by two slots, then  $\alpha = 2\beta$  and so on.



## EMF Equation:-

Consider a 3Ø alternator with “p” no. of poles driven at a constant speed N rpm.

Let  $E_p$  be the rms values of the induced emf per pole.

Ø is the average flux per pole in Webbers

Z is the no. of stator conductor per phase

T is the no. of stator turns per phase and we know  $T = Z/2$

F is the frequency of induced emf in hz

Therefore total flux cut per revolution by any one stator conductor is equal to  $PØ$  Webbers.

Time taken for one revolution is equal to  $1/N$  min or  $60/N$  sec

Therefore rate of cutting of flux is equal to  $= dØ/dt$

$$= “PØ” / “60/N” \text{ (wb/sec)}$$

$$= PØN/60 \text{ (wb/sec)}$$

But we have  $F = PN/120$

Therefore  $2F = PN/60$

$$d\phi/dt = 2\phi F$$

According to faraday's second law of EMI

➤ The average value of the induced emf =  $2F\phi$  volts

➤ In any one phase  $Z$  conductors are joined in series, therefore average induced emf per phase =  $2F\phi Z$  volts

$$= 2F\phi(2T) \text{ volts}$$

Avg. value of induced emf per phase =  $4F\phi T$  volts

For a sine wave form factor = rms value/avg value = 1.11

RMS value of induced emf per phase =  $1.11 \times 4F\phi T$

$$= 4.44F\phi T \text{ volts}$$

➤ In a practical alternator the space distribution of the filed flux is not purely sinusoidal, it is having some distortion and moreover in a practical alternator short pitch winding is used, therefore by these two reasons , the actual EMF that is induced is somewhat less than the emf that is arrived at.

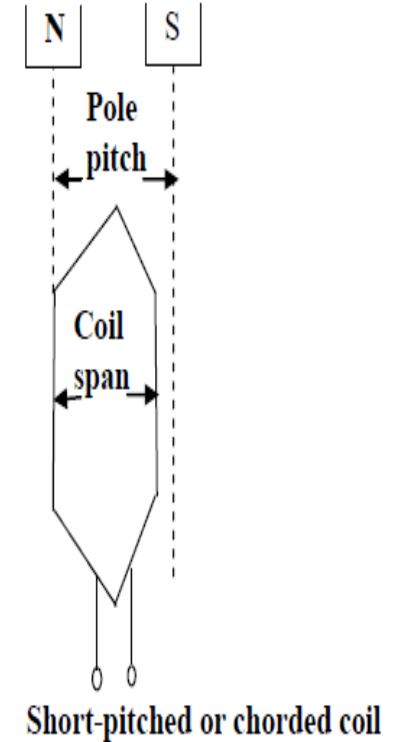
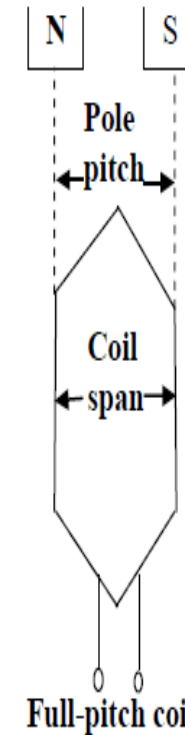
➤ Therefore by inscrting pitch factor (or) chording factor (or) coil span factor (  $K_c$  or  $K_p$ ) and Distribution or breadth factor ( $K_d$  or  $K_b$ ) in the above emf equation , we with get the actual emf equations as  $E = 4.44 f \Phi T K_c K_d$  volts .

### Pitch factor (or) chording factor (or) coil span factor :-

It is the ratio of vector sum of the emfs induced in the two coil sides of coil to their arithmetic sum .

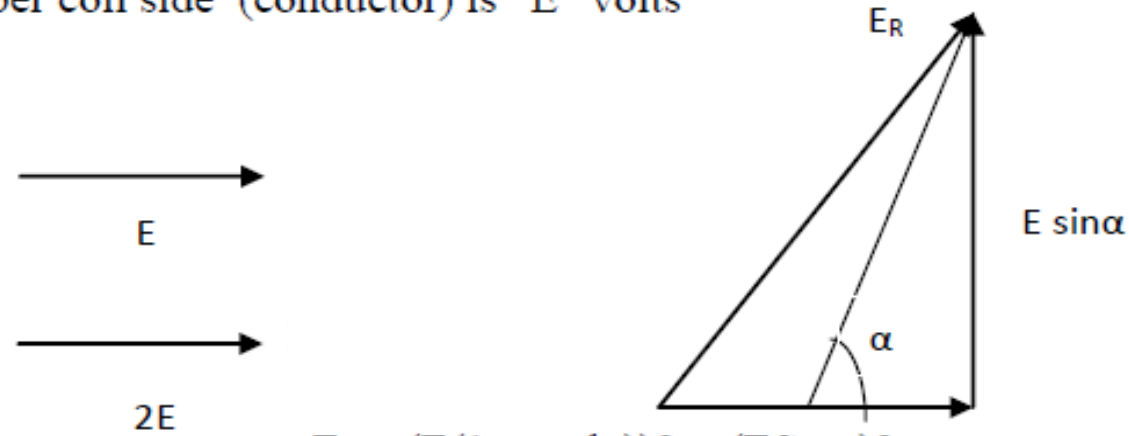
$K_c = \text{vector sum of induced emfs per coil} / \text{arithmetic sum of induced emf per coil}$

$= \text{voltage induced in short pitch winding} / \text{voltage induced in full pitch winding}$



Let the coil span of the short pitch winding is less than one pole pitch ( $180^\circ$ ) by an angle  $\alpha$ .

Let the emf induced per coil side (conductor) is “E” volts



$$E_r = (E(1+\cos\alpha))^2 + (E\sin\alpha)^2$$

$$= E^2 (1+1+2\cos\alpha)$$

$$= E^2 (2+2\cos\alpha/2)$$

$$= 2E^2 \cos\alpha/2$$

Arithmetic sum of the EMF's around, the coil is equal to  $E+E=2E$  volts. Vector sum of the EMF's around the coil in short pitch winding is equal

$$OA+AB=OB$$

$$OA+AB=OB$$

$$2.OA.\cos(\alpha/2)$$

$$K_c = \text{vector} / \text{arithmetic sum} = 2E\cos(\alpha/2) / 2E = \cos(\alpha/2)$$

Therefore  $K_c = \cos(\alpha/2)$  { Where  $\alpha$  is the short pitch angle (or)  $\lambda$  }

### **Distributions Factor (or) breadth factor:- ( $K_d$ or $K_b$ )**

- When the coils comprising a phase of the windings are distributed in two or more slots per pole the e.m.fs in the adjacent coils will be out of phase with respect to one another and their resultant will be less than their algebraic sum.
- The ratio of the vector sum of the e.m.fs induced in all the coils distributed in a number of slots under one pole to the arithmetic sum of the e.m.fs induced (or to the resultant of the e.m.fs induced in all the coils concentrated in one slot under one pole) is known as distributed factor  $K_d$ .
- It is the ratio of voltage induced in a distribution winding to the voltage induced in the concentric winding.
- Let 'm' be the no. of stator slots per pole per phase.

$B = 180^\circ / \text{no. of slots/pole}$  Which is known as slot angle.

E is the EMF induced per conductor. With radius 'R' and cent 'O' a circle is drawn to pass through the points ABCD

Arithmetic sum of EMF'S =  $AB + BC + CD + \dots + DF$  .....

=  $E + E + E + E + E$  ..... m times

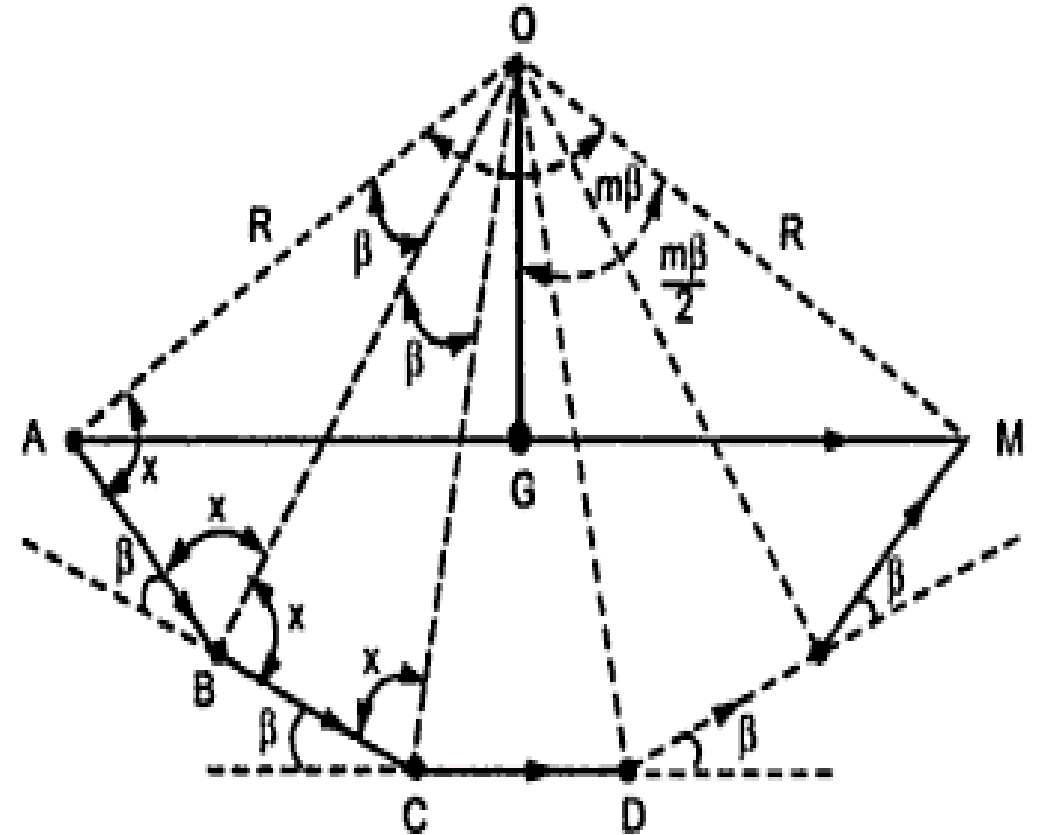
= m E

From fig. vector sum of the EMF'S =  $AF = 2AS$

From the  $\Delta AOS$ ,  $\sin(m\beta/2) = AS/OA = AS/R$

$AS = R \sin(m\beta/2)$

Vector sum =  $AF = 2AS = 2R \sin(m\beta/2)$



From the  $\Delta AOG$   $\sin(\beta/2) = AG/OA = AG/R = E/2R$

$$E = 2R \sin(\beta/2)$$

$$\text{Vector sum} = AF = 2AS = 2R \sin(m\beta/2)$$

From the  $\Delta AOG$   $\sin(\beta/2) = AG/OA = AG/R = E/2R$

$$E = 2R \sin(\beta/2)$$

$$\begin{aligned} \text{There } K_d &= \text{vector sum} / \text{arithmetic sum} = 2R \sin(m\beta/2) / mE \\ &= 2R \sin(m\beta/2) / m \cdot 2R \sin(\beta/2) \end{aligned}$$

$$\text{Therefore } K_d \text{ or } K_b = \sin(m\beta/2) / \sin(\beta/2)$$

$$\text{Also } K_d \text{ or } K_b = \sin(mn\beta/2) / m \sin(n\beta/2) \quad \{\text{where } n = \text{order of harmonic}\}$$

## **Concentration and Distribution Windings:**

- Each coil side contains a no. of conductors, if all the conductors of a coil side are placed in a single slot, it is called concentrated coil.
- It gives more voltage but the sine wave will not be smooth.
- When the conductors of the coil side are distributed in diff. slots, it is called as distributed slots.
- It gives less voltage but the wave form will be smooth.

### **Advantages of distributed windings:-**

- The generated voltage waveform will be improved, is more sinusoidal. The magnitude of induced voltage will be reduced

## Short-Pitch Winding

### Advantages:-

- Copper in end connection can be saved.
- Harmonics are reduced
- Iron losses will be reduced
- Efficiency will be increased
- Generated voltage waveform will be improved is more sinusoidal.

### Disadvantages:-

- The magnitude of the induced voltage will be reduced